

Deliverable 3: Use-Case Flow Specification

Course: Software Engineering Lab 1 (PES University, Dept. of CSE)

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Problem Statement #05: Academic Elective Bidding & Allocation System

Use Case: UC-01 — Submit Elective Bids

Primary Actor: Student | **Secondary Actor:** Academic Database | **Priority:** High

Preconditions:

- Student is authenticated and in good academic standing.
- Elective bidding window is active/open.
- Student has exactly 100 unspent bidding points available.

Postconditions:

- Student's elective bids and credit allocations are persistently stored in the database.
- Total credits used are marked as 100.
- An immutable audit log entry and confirmation receipt are generated.

Main Success Scenario

1. Student logs into portal and navigates to "Elective Course Bidding" section.
2. System displays offered elective courses, available seats, faculty, and slot timings.
3. Student selects electives and distributes exactly 100 bidding points among them based on priority.
4. Student submits bidding form by clicking "Submit Bids".
5. System executes **UC-02: Validate Course Prerequisites** to verify completed credits against prerequisites.
6. System validates that sum of allocated bidding points equals exactly 100.
7. System confirms that no two selected electives share mutually exclusive core slots.
8. System commits bid records into active allocation pool and generates transaction timestamp.
9. System displays confirmation message and locked preference summary.
10. Use Case ends successfully.